

Path integral molecular dynamics: A high-fidelity approach to quantum dynamics of electrons

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ABSTRACT

We investigate electron transport in the uniform electron gas using ring-polymer molecular dynamics (RPMD). Working in the weakly coupled, non-degenerate regime, we use RPMD to probe how the onset of quantum diffraction effects at high temperature reshapes electron-electron collisions and leads to a classical-to-quantum crossover in macroscopic transport properties. Static thermodynamics obtained with RPMD are consistent with the weak-coupling equation of state, confirming correct quantum Boltzmann sampling. Real-time transport extracted from mean square displacements exhibits the expected ballistic-to-diffusive transition and a systematic reduction of the electronic self-diffusivity as quantum effects strengthen, due to quantum diffraction modifying electron-electron collisions. Direct ring-polymer scattering simulations reveal diffractive “softening” of binary deflections, providing a micro-to-macro link between collision physics and diffusion. The present study establishes RPMD as a quantitative, trajectory-based tool for electron transport across the classical-quantum crossover and furnishes benchmarks for improving Coulomb-log interpolation models. We outline extensions to multi-component plasmas and a path to incorporate Fermi-Dirac statistics within path-integral dynamics.

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I. INTRODUCTION

Understanding electron transport in plasmas is central to high-energy-density (HED) physics, astrophysics, and fusion science. Diffusion, conductivity, and temperature relaxation are sensitive to how binary collisions are modified by Coulomb correlations, quantum diffraction, and exchange.^{1–3} As a minimal, widely used model, we consider the *uniform electron gas* (UEG; also called the electron one-component plasma), in which electrons move in a uniform neutralizing background.¹ With density n and temperature T as the only control parameters, the UEG is an ideal platform to examine this classical-to-quantum scattering crossover.⁴ This crossover is where close collisions transition from Rutherford-like scattering to diffraction-dominated quantum scattering. It occurs when the electron thermal de Broglie wavelength becomes comparable to the classical scattering length, which for typical warm-dense-matter densities corresponds to temperatures of a few to tens of electronvolts, a regime that remains difficult to access with first-principles simulations and is sparsely benchmarked.⁵

In weakly coupled, non-degenerate plasmas, conventional Coulomb-logarithm-based closures work well in the classical limit but

lose systematic control across the classical-to-quantum crossover.^{6,7} We define the electron-electron Coulomb logarithm as $\ln \Lambda = \ln(b_{\max}/b_{\min})$. The upper limit $b_{\max}$ is taken as the distance beyond which collective Debye shielding renders binary encounters ineffective, while the lower limit $b_{\min}$ corresponds to the thermal distance of closest approach below which electrons cannot penetrate in classical scattering.⁸

The *NRL Plasma Formulary* provides a smooth quantum transition for the electron-electron Coulomb logarithm $\ln \Lambda$, while noting truncation errors that scale as $\mathcal{O}(1/\ln \Lambda)$ and for $\ln \Lambda \lesssim 2$ become order unity.⁹ Classical point-particle MD provides a robust baseline for the classical scattering regime; however, it omits diffraction, and purely classical closures lose accuracy across this crossover. Hence, simulation benchmarks that remain valid as diffraction emerges are needed; classical MD, though reliable in the classical limit, cannot resolve the classical-quantum crossover. To interrogate the crossover with fully interacting trajectories that include quantum delocalization, we employ ring-polymer molecular dynamics (RPMD).^{10–12}

RPMD maps each quantum particle to a classical ring polymer of P beads, yielding exact quantum equilibrium as $P \rightarrow \infty$ and the

classical limit at $P = 1$.^{11,12} RPMD has been widely applied in chemical physics to capture nuclear quantum effects (NQE)—zero-point energy and tunneling—in dynamical observables,¹¹ but its performance for electron transport in plasmas, where long-range Coulomb forces dominate, has not been explored to the best of our knowledge. RPMD provides a controlled, trajectory-based way to isolate diffraction-driven modifications to scattering and to benchmark kinetic theory predictions, even when the Coulomb logarithm is poorly constrained.

In HED conditions, path-integral molecular dynamics (PIMD) has most commonly been applied to nuclei to capture nuclear quantum effects (NQE), and it has proven highly effective for low-temperature ions. This motivates using RPMD for high-temperature electrons: since the thermal de Broglie wavelength $\Lambda_{th} \propto (mT)^{-1/2}$, the electron temperature needed to reach the proton's quantum delocalization scale is higher by the proton-electron mass ratio, $T_e \approx (m_p/m_e) T_i \approx 1836 T_i$. At 10 g cm^{-3} and $T_i = 0.1 - 1 \text{ eV}$, including NQEs via PIMD increases self-diffusion by $\sim 20\% - 146\%$ and reduces electrical/thermal conductivities relative to classical nuclei.¹³ Furthermore, modern first-principles equation-of-state (EOS) work now treats ions quantum-mechanically via PIMD across broad $\rho - T$ ranges.¹⁴ Thus $T_i = 0.1 - 1 \text{ eV}$ maps to $T_e \approx 1.7 \times 10^2 - 1.7 \times 10^3 \text{ eV}$, providing a quantitative rationale for our use of RPMD to probe electron transport in that regime.

We note that path-integral approaches have also been used to model finite-temperature electron plasmas with explicit Fermi statistics, most notably in the restricted path-integral molecular dynamics (RPIMD) work of Oh and Deymier.^{15,16} Because of the computational cost associated with enforcing the restricted paths and evaluating determinants along the imaginary-time chains, those simulations were necessarily limited to relatively small systems and short trajectories. By contrast, the present work focuses on Boltzmann, non-degenerate electrons ($n\Lambda_{th}^3 \ll 1$) and employs ring-polymer molecular dynamics with the bare Coulomb interaction to access larger systems and long trajectories, with the goal of characterizing how quantum diffraction modifies near-equilibrium electron-electron scattering and transport.

To set the stage, we adopt standard notation for the uniform electron gas (UEG) and define the parameters that govern the classical-quantum scattering crossover.¹⁷ Classical coupling is set by $\Gamma \equiv q^2 / (4\pi\epsilon_0 a k_B T)$, where $a = (3/4\pi n)^{1/3}$ is the Wigner-Seitz radius, n the electron number density, and T the temperature; weak coupling corresponds to $\Gamma \ll 1$, where Coulomb forces can be treated perturbatively, while $\Gamma \gtrsim 1$ indicates strong coupling with non-perturbative correlations.¹

Quantum degeneracy is measured by $\Theta \equiv T/T_F$, with Fermi temperature $T_F = \hbar^2 (3\pi^2 n)^{2/3} / (2m_e k_B)$; $\Theta \ll 1$ denotes a degenerate electron gas, $\Theta \sim 1$ the crossover, and $\Theta \gg 1$ the classical limit.^{1,18} The thermal de Broglie wavelength $\Lambda_{th} \equiv h / \sqrt{2\pi m_e k_B T}$ sets the spatial extent of electron wave packets; when $n\Lambda_{th}^3 \gtrsim 1$, neighboring wavefunctions overlap and quantum diffraction/exchange become unavoidable, whereas $n\Lambda_{th}^3 \ll 1$ indicates point-particle behavior. These measures are linked by $\Theta \propto (n\Lambda_{th}^3)^{-2/3}$, so that $\Theta \ll 1$ and $n\Lambda_{th}^3 \gg 1$ are equivalent markers of strong degeneracy.²

To parameterize the classical-quantum crossover, we introduce $\zeta \equiv b_Q/b_{Cl}$, choosing $b_Q = \Lambda_{th}$ the thermal de Broglie wavelength—beyond which wavepacket overlap and diffraction enter—and $b_{Cl} = e^2 / (4\pi\epsilon_0 k_B T)$ the classical distance of closest approach for a

head-on collision at thermal energy (the impact parameter where the Coulomb potential equals $k_B T$); collisions follow classical Rutherford-like behavior for $\zeta \ll 1$ and become increasingly quantum for $\zeta \gtrsim 1$ (diffraction dominated).² Figure 1 summarizes these regimes in the T - n plane.

In this work, we use RPMD to study electron transport in the UEG at fixed weak coupling ($\Gamma = 0.1$). To avoid the fermion sign problem in path-integral treatments of electrons, we restrict our attention to the non-degenerate regime ($\Theta \gg 1$), where exchange is negligible and Boltzmann statistics apply.¹⁹ We probe the classical-to-quantum scattering crossover by varying ζ and report three main findings:

1. Static thermodynamics: RPMD reproduces the expected weak-coupling UEG equation of state to percent-level accuracy, confirming correct Boltzmann sampling.^{20–22}
2. Transport: Diffusion coefficients extracted from RPMD trajectories match classical expectations at small ζ and trends toward quantum-diffraction behavior as ζ grows, delineating a crossover where asymptotic formulas are least reliable.^{6,7,9}
3. Microscopic mechanism: Ensemble ring-polymer scattering simulations show a clear softening of binary deflections for $b \lesssim b_Q$, directly linking diffractive collisions to the observed reduction in diffusion at large ζ .

These results motivate future extensions of the method to *multi-component plasmas* (e.g., hydrogenic mixtures), where electron-ion and ion-ion processes can be assessed alongside emerging routes to incorporate Fermi-Dirac statistics in path-integral dynamics.^{6,7,23}

We emphasize that the present work is confined to near-equilibrium electron transport in the weakly coupled, non-degenerate UEG. We do not attempt to describe any high frequency transport processes or processes that are far from equilibrium. Our focus is

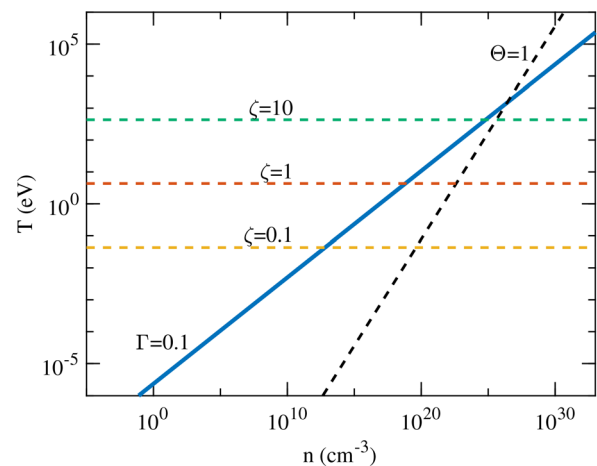


FIG. 1. Phase diagram in the T - n plane. The black diagonal dashed line marks $\Theta = 1$, separating non-degenerate ($\Theta > 1$, upper left) from degenerate ($\Theta < 1$, lower right) electrons. The solid blue curve shows $\Gamma = 0.1$, demarcating a weakly coupled region above, from more strongly coupled conditions below. Horizontal dashed lines indicate constant $\zeta = b_Q/b_{Cl}$ with $b_Q = \Lambda_{th}$ and $b_{Cl} = e^2 / (4\pi\epsilon_0 k_B T)$: $\zeta \ll 1$ corresponds to classical Rutherford scattering, while $\zeta \gtrsim 1$ signals diffraction/exchange-dominated collisions where a quantum treatment is required.

instead on how quantum diffraction modifies low-frequency, near-equilibrium transport coefficients.

II. MODEL AND METHODS

A. Ring-polymer molecular dynamics framework for electron transport

The quantum delocalization of electrons is described by Feynman's imaginary-time path integral for the canonical partition function,^{12,24}

$$Z = \text{Tr}[e^{-\beta\hat{H}}], \quad \hat{H} \equiv \hat{T} + \hat{V},$$

$$\hat{T} = \sum_{j=1}^N \frac{\hat{\mathbf{p}}_j^2}{2m_j}, \quad \hat{V} = U(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (1)$$

Here $\beta \equiv 1/(k_B T)$. Using the second-order symmetric Suzuki-Trotter factorization,^{12,24}

$$e^{-\beta\hat{H}} \approx \left(e^{-\frac{\beta}{2P}\hat{V}} e^{-\frac{\beta}{P}\hat{T}} e^{-\frac{\beta}{2P}\hat{V}} \right)^P + \mathcal{O}\left(\frac{1}{P^2}\right), \quad (2)$$

the Boltzmann operator is expressed as a product of P short-time factors with imaginary-time step $\delta\beta \equiv \beta/P$.

To make these factors explicit, we insert P copies of the identity in the position basis, $\mathbf{1} = \int d\mathbf{r} |\mathbf{r}\rangle\langle\mathbf{r}|$, between them and use the short imaginary time free-particle matrix element (the thermal kernel),

$$\langle\mathbf{r}|e^{-\varepsilon\hat{T}}|\mathbf{r}'\rangle = \left(\frac{m}{2\pi\hbar^2\varepsilon}\right)^{3/2} \exp\left[-\frac{m\|\mathbf{r}-\mathbf{r}'\|^2}{2\hbar^2\varepsilon}\right], \quad (3)$$

for a particle of mass m ; in the many-body case the kinetic factor factorizes over particles j with masses m_j . These steps turn the trace into a classical integral over P replicas ("beads") of each particle,

$$Z \approx \int \prod_{i=1}^P d\mathbf{r}^{(i)} \exp\left[-\beta \sum_{i=1}^P \left(\sum_{j=1}^N \frac{m_j \omega_P^2}{2} \|\mathbf{r}_j^{(i)} - \mathbf{r}_j^{(i+1)}\|^2 + \frac{1}{P} U(\mathbf{r}_1^{(i)}, \dots, \mathbf{r}_N^{(i)}) \right)\right], \quad (4)$$

with cyclic indexing $\mathbf{r}_j^{(P+1)} \equiv \mathbf{r}_j^{(1)}$ and $\omega_P = P/(\beta\hbar)$. The Gaussian kinetic matrix elements generate the harmonic "springs" between neighboring beads, and each potential factor contributes U/P evaluated beadwise. A schematic is shown in Fig. 2: the left icon corresponds to the classical limit $P = 1$, the center shows a ring polymer with

$P = 8$ harmonically coupled beads, and the right cloud illustrates the quantum particle's equilibrium distribution in the $P \rightarrow \infty$ limit. In this limit, the ring-polymer isomorphism becomes exact.

To enable molecular-dynamics sampling, introduce auxiliary classical momenta for each bead and particle via the Gaussian identity,

$$1 = \prod_{i=1}^P \prod_{j=1}^N \left(\frac{\beta}{2\pi m_j} \right)^{3/2} \int d\mathbf{p}_j^{(i)} \exp\left[-\beta \frac{\|\mathbf{p}_j^{(i)}\|^2}{2m_j}\right]. \quad (5)$$

Combining Eqs. (4) and (5) gives the *phase-space* classical isomorphism,

$$Z = \int \prod_{i=1}^P d\mathbf{r}^{(i)} d\mathbf{p}^{(i)} \exp\left[-\beta H_P(\{\mathbf{r}^{(i)}, \mathbf{p}^{(i)}\})\right], \quad (6)$$

with ring-polymer Hamiltonian

$$H_P = \sum_{i=1}^P \left[\sum_{j=1}^N \frac{\|\mathbf{p}_j^{(i)}\|^2}{2m_j} + \frac{m_j \omega_P^2}{2} \|\mathbf{r}_j^{(i)} - \mathbf{r}_j^{(i+1)}\|^2 + \frac{1}{P} U(\mathbf{r}_1^{(i)}, \dots, \mathbf{r}_N^{(i)}) \right]. \quad (7)$$

Here, $\mathbf{r}_j^{(i)}$ denotes the bead i of particle j and $\mathbf{p}_j^{(i)}$ its conjugate momentum; the cyclic condition $\mathbf{r}_j^{(P+1)} \equiv \mathbf{r}_j^{(1)}$ closes each ring polymer. The spring term encodes the quantum kinetic operator $\hat{T}$, while the bead average encodes $\hat{V}$; sampling $e^{-\beta H_P}$ recovers the Trotter-discretized quantum equilibrium in the limit $P \rightarrow \infty$.^{11,12} Setting $P = 1$ in (7) recovers the classical Hamiltonian as $\mathbf{r}_j^{(2)} \equiv \mathbf{r}_j^{(1)}$ and the bead average potential reduces to U ,

$$H_{P=1} = \sum_{j=1}^N \frac{\|\mathbf{p}_j\|^2}{2m_j} + U(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (8)$$

In RPMD, the ring-polymer Hamiltonian H_P generates classical equations of motion in the extended phase space $\{\mathbf{r}^{(i)}, \mathbf{p}^{(i)}\}_{i=1}^P$. Correlation functions evaluated along these trajectories approximate the *Kubo-transformed* quantum time-correlation function, which are equivalent in linear-response theory. Following Habershon *et al.*,¹¹ the Kubo transform of two operators A and B is

$$\tilde{C}_{AB}(t) = \frac{1}{\beta Z} \int_0^\beta d\lambda \text{Tr}\left[e^{-(\beta-\lambda)\hat{H}} \hat{A}(0) e^{-\lambda\hat{H}} \hat{B}(t)\right], \quad (9)$$

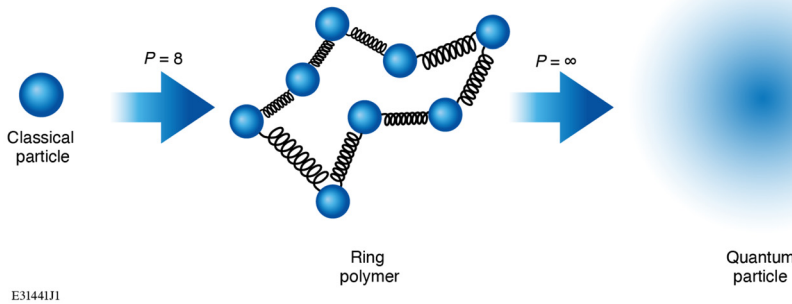


FIG. 2. Ring-polymer isomorphism: a quantum particle is represented by a classical ring polymer of P harmonically coupled beads (center; shown for $P = 8$). Equilibrium properties are exact as $P \rightarrow \infty$ (right); $P = 1$ yields a classical point particle (left). Illustration inspired by Unke *et al.*²⁵

where $Z = \text{Tr}[e^{-\beta\hat{H}}]$ is the quantum partition function and $B(t) = e^{i\hat{H}t/\hbar} B e^{-i\hat{H}t/\hbar}$ is the *real-time* Heisenberg operator. The Kubo transform is real, symmetric, finite at short times, and reduces to the classical time-correlation function in the limit $\beta\hbar \rightarrow 0$. In the corresponding low-frequency limit relevant here, the ring-polymer spectral density provides an accurate approximation to the Kubo spectrum,

$$C_{AB}^{\text{RP}}(\omega) \rightarrow \tilde{C}_{AB}(\omega) \quad \text{as } |\omega| \rightarrow 0, \quad (10)$$

which is the focus of this study.

RPMD provides an approximate, but formally well-defined, classical mechanism for evaluating $\tilde{C}_{AB}(t)$ directly from ring-polymer trajectories.^{10,11} It is precisely this ability to generate approximate pseudo-real-time dynamics for fully interacting many-electron systems that distinguishes RPMD from imaginary-time approaches.

The pseudo-real-time dynamics of RPMD are reliable only in the *low-frequency* sector of the Kubo-transformed spectrum. As emphasized by Craig and Manolopoulos¹⁰ and by Habershon *et al.*,¹¹ physical frequencies must lie below the first internal ring-polymer (Matsubara) mode,

$$\omega_{\text{RP}} = \frac{2\pi}{\beta\hbar} = \frac{2\pi k_B T}{\hbar}, \quad (11)$$

above which spurious internal vibrations contaminate the dynamics. For the uniform electron gas studied here, the relevant collective scale is the plasma frequency $\omega_p = \sqrt{ne^2/(\epsilon_0 m_e)}$. In all our simulations we find $\omega_p/\omega_{\text{RP}} \ll 1$ (see Table II), ensuring that the transport properties we compute lie safely inside the low-frequency regime where RPMD accurately approximates Kubo-transformed correlation functions.^{10–12} Thus, no nonequilibrium extension of the method is invoked; only the linear-response, near-equilibrium domain for which RPMD was designed.

In contrast with RPMD's approximate pseudo-real-time dynamics, Path-Integral Monte Carlo (PIMC) is the gold standard for *static* equilibrium properties: it is a fully *ab initio* method rooted in Feynman's imaginary-time path integrals²⁴ and samples the full N -electron quantum statistics without mean-field approximations.²⁶ However, PIMC accesses only imaginary-time correlation functions, and recovering real-time behavior from such data requires analytic continuation—an exponentially ill-conditioned inverse problem. Even with modern stochastic continuation techniques,^{27,28} reliable extraction of low-frequency dynamical information remains extremely challenging. RPMD circumvents this difficulty by providing a practical approximation to real-time Kubo-transformed correlation functions, enabling direct estimation of dynamical observables such as the diffusion coefficient without analytic continuation.

A separate family of approaches modifies classical interactions via *quantum statistical potentials* (QSPs), adding static, temperature-dependent terms to mimic diffraction and Pauli exclusion. As emphasized by Graziani *et al.*, QSPs treat all pairs with the same average interaction (ignoring relative velocity), neglect wavefunction deformation during close encounters, and enforce Pauli blocking through coordinate-space exclusion rather than proper momentum-space antisymmetry.²⁹ These limitations can yield qualitatively incorrect dynamics in dense plasmas, prompting more sophisticated momentum-dependent or wave-packet-based schemes.^{30,31}

An alternative, explicitly wave-mechanical route is *wave-packet molecular dynamics* (WPMD), which represents each electron as a

Gaussian wave packet evolving under simplified time-dependent Schrödinger dynamics, with ions treated classically.^{32,33} In its basic form, each electron is a single Gaussian whose centroid and width evolve in time; many-body antisymmetry can be imposed via a Slater determinant or approximated with Pauli potentials.³⁴ Recent wave-packet electron-dynamics formulations have been used to compute dynamic structure factors in warm dense hydrogen without invoking the Born–Oppenheimer approximation or the Chihara decomposition.³⁵ Semi-empirical extensions such as the electron force field (eFF) introduce scaled Pauli and correlation terms that can match stopping powers and equations of state for warm dense hydrogen,^{36,37} but their parameters are not *a priori* predictive and require validation against experiment or higher-level theory.^{31,32}

Another prominent class of simulations is *ab initio* molecular dynamics based on density functional theory (DFT), combined with the Kubo–Greenwood (KG) formalism to compute frequency-domain conductivities.³⁸ This approach delivers reliable transport in many WDM settings,^{38–43} but electron–electron correlations enter only via approximate exchange–correlation functionals, so many-body effects are limited by the chosen functional.^{44–46} In the same mean-field spirit, time-dependent DFT (TD-DFT) propagates Kohn–Sham orbitals in real time to obtain dynamical response,^{47,48} yet remains constrained by the exchange–correlation approximation (e.g., adiabatic/local forms), which can miss genuine many-body dynamics. For orientation, Table I contrasts common approaches to electron transport by their dynamics domain and treatment of electron–electron interactions.

B. Computational details

We restrict ourselves to the weakly coupled ($\Gamma \ll 1$), non-degenerate ($\Theta \gg 1$) regime with the Coulomb coupling fixed at

TABLE I. Comparison of simulation methods for electron transport.

Method	Time domain	Electron interaction level
RPMD	Pseudo-real	Full
PIMC	Imaginary	Full
DFT+KG	Fourier space	Approx. XC (static)
TD-DFT	Real	Approx. XC (dynamic)
QSP-MD	Real	Model-dependent
WPMD	Real	Semi-empirical

TABLE II. Simulation parameters for $\Gamma = 0.1$ over varying $\zeta = b_0/b_{\text{Cl}}$.

ζ	T (eV)	n (cm ⁻³)	$n\Lambda_{ih}^3$	$\omega_p/\omega_{\text{RP}}$
0.01	4.40×10^{-4}	6.7×10^6	2.4×10^{-10}	3.48×10^{-5}
0.10	4.40×10^{-2}	6.9×10^{12}	2.5×10^{-7}	3.53×10^{-4}
1	4.33×10^0	6.5×10^{18}	2.4×10^{-4}	3.48×10^{-3}
2	1.72×10^1	4.1×10^{20}	1.9×10^{-3}	6.96×10^{-3}
3	3.98×10^1	5.0×10^{21}	6.7×10^{-3}	1.05×10^{-2}
5	1.103×10^2	1.1×10^{23}	3.1×10^{-2}	1.78×10^{-2}
7	2.132×10^2	7.7×10^{23}	8.2×10^{-2}	2.43×10^{-2}
10	4.246×10^2	6.1×10^{24}	2.3×10^{-1}	3.44×10^{-2}

$\Gamma = 0.1$, while the quantum-classical length-scale ratio $\zeta = b_Q/b_{Cl}$ is varied between 0.01 and 10. Table II lists the corresponding temperature T , number density n , and quantum concentration $n\Lambda_{th}^3$; in every case $n\Lambda_{th}^3 \ll 1$ ensures non-degeneracy, and $\zeta \geq 1$ marks where quantum diffraction is expected to become significant. To quantify the dynamical regime of the simulations, Table II also lists the ratio ω_p/ω_{RP} , which remains $\ll 1$ across all cases, confirming that the simulated dynamics lie well within the low-frequency regime where RPMD is reliable.

All ring-polymer MD simulations were performed with i-PI 3.0,⁴⁹ using its client-server socket interface with LAMMPS as the force engine.^{50,51} We simulated $N = 5000$ electrons in a cubic cell with periodic boundary conditions. Long-range Coulomb interactions were computed with PPPM on a $64 \times 64 \times 64$ mesh, targeting a relative RMS force error of 10^{-6} . Equilibration used the *path-integral Langevin equation* thermostat in its *global* variant (PILE-G), which thermostats the ring-polymer normal modes globally.⁵² The path integral was discretized with $P = 10 - 18$ beads (with $P = 1$ runs used as classical baselines). Production lengths were $N_{steps} = 10^6$ MD steps for all ζ .

The time step was set by the case-dependent plasma frequency ω_p with

$$\Delta t = 5 \times 10^{-4} \omega_p^{-1}. \quad (12)$$

Using Eq. (12) and $T_p \equiv 2\pi/\omega_p$, one period spans $T_p/\Delta t = 2\pi/(5 \times 10^{-4}) \approx 1.26 \times 10^4$ steps; thus a 10^6 -step production corresponds to $t_{tot} \approx 79.6 T_p$ (i.e., about 79.6 plasma periods).

We employ the second-order Trotter (primitive) factorization with $P = 10$ for $\zeta = 0.1 - 3$, $P = 14$ for $\zeta = 5$ and 7 , and $P = 18$ for $\zeta = 10$. Convergence of static observables with respect to P was verified; the residual Trotter error scales as $\mathcal{O}(1/P^2)$ and remained below our statistical uncertainties.^{12,19}

For $P > 1$, transport properties are computed from the ring-polymer *centroid* trajectories; the centroid is the quasi-classical coordinate in RPMD and equals the particle position in the classical limit $P = 1$.^{10,11} In our data, mean-square displacements (MSDs) and diffusivities obtained from all-bead averages were statistically indistinguishable from the centroid-only results, so we report the latter for simplicity and I/O efficiency.

III. RESULTS AND DISCUSSION

A. Equation of state

We report the excess potential energy per particle normalized by thermal energy, $U_{ex}/(Nk_B T)$, evaluated as a function of ζ at fixed Coulomb coupling $\Gamma = 0.1$. For each state point, U_{ex} is averaged over the production window and normalized by $Nk_B T$. Figure 3 illustrates $U_{ex}/(Nk_B T)$ for classical ($P = 1$) and ring-polymer ($P = 10, 14$, and 18) cases across the ζ values in Table II, enabling a convergence check of static properties with bead number. At $\zeta = 5$ and 7 , $P = 14$ agrees with $P = 10$ within the error bars, indicating convergence by $P = 14$. At $\zeta = 10$, $P = 14$ lies above $P = 10$ yet remains within the $P = 18$ uncertainty, implying at least $P \geq 14$ is required there.

As a reference, we plot the weak-coupling analytic UEG equation of state (EOS) (Abe-Cohen-Murphy expansion, through $\mathcal{O}(\Gamma^{9/2})$),

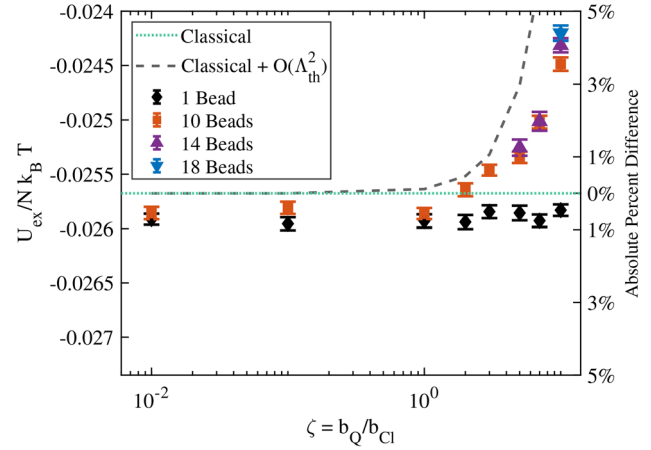


FIG. 3. Normalized potential energy $U_{ex}/(Nk_B T)$ vs $\zeta = b_Q/b_{Cl}$ at $\Gamma = 0.1$. Black diamonds: $P = 1$ (MD); red squares: $P = 10$ (RPMD); purple triangles: $P = 14$ (RPMD); blue inverted triangles: $P = 18$ (RPMD). Green dotted: Abe-Cohen-Murphy weak-coupling EOS; black dashed: Wigner-Kirkwood leading $\mathcal{O}(\Lambda_{th}^2)$ correction added to Abe-Cohen-Murphy. The right ordinate gives the absolute percent deviation relative to the classical Abe-Cohen-Murphy curve (green dotted).

$$\frac{U_{ex}}{Nk_B T} = -\frac{\epsilon}{2} - \left(\frac{\epsilon}{2}\right)^2 \left[\ln \epsilon + \ln 3 + 2\gamma_E - \frac{4}{3} \right] - \left(\frac{\epsilon}{2}\right)^3 [3 \ln \epsilon - c] \quad (13)$$

$$\epsilon = \sqrt{3} \Gamma^{3/2}, \quad (14)$$

with Euler-Mascheroni constant γ_E and $c \simeq 2.0224$.^{20–22} At fixed Γ , this classical EOS is independent of ζ because the configurational part of the classical partition function is mass-independent. Thus $U_{ex}/(Nk_B T)$ depends only on Γ , and any ζ -dependence in the data reflects quantum effects via Λ_{th} .

To quantify quantum diffraction beyond the classical Abe-Cohen-Murphy EOS, we include the leading Wigner-Kirkwood $\mathcal{O}(\hbar^2) \sim \mathcal{O}(\Lambda_{th}^2)$ correction derived for the OCP.^{53,54} Re-expressed in our (Γ, ζ) variables with $\zeta \equiv b_Q/b_{Cl}$ (where $b_Q \equiv \Lambda_{th}$ and $b_{Cl} \equiv e^2/4\pi\epsilon_0 k_B T$), the $\mathcal{O}(\Lambda_{th}^2)$ contribution to the reduced excess internal energy is

$$\left. \frac{U_{ex}}{Nk_B T} \right|_{\mathcal{O}(\Lambda_{th}^2)} = \frac{\Gamma^3 \zeta^2}{8\pi} + \mathcal{O}(\Lambda_{th}^4). \quad (15)$$

We therefore plot the classical Abe-Cohen-Murphy EOS augmented by this leading quantum term,

$$\frac{U_{ex}}{Nk_B T} = \left(\frac{U_{ex}}{Nk_B T} \right)_{Abe/CM} + \frac{\Gamma^3 \zeta^2}{8\pi}. \quad (16)$$

This $\mathcal{O}(\Lambda_{th}^2)$ correction is structure-independent (it follows from Poisson's equation for the Coulomb OCP) and is valid when $\zeta \ll 1$ (weak diffraction) under weak coupling; when $b_Q \gtrsim b_{Cl}$ (i.e., $\zeta \gtrsim 1$), the Wigner-Kirkwood expansion ceases to converge and non-perturbative treatments are required.²²

For $P = 1$, the absolute percent deviation from the classical analytic value is $\leq 1.09\%$ across all eight state points. For the ring-polymer

data ($P = 10\text{--}18$), when compared to the classical EOS augmented by the leading $O(\Lambda_{\text{th}}^2)$ quantum correction, the first five state points ($\zeta = 0.01\text{--}3$) agree to within 1%, while the deviation grows systematically for larger ζ (about 2.68% at $\zeta = 5$, 5.45% at $\zeta = 7$, and 12.07% at $\zeta = 10$). These trends are consistent with the expected breakdown of the $O(\Lambda_{\text{th}}^2)$ (Wigner–Kirkwood) approximation as $\zeta = b_Q/b_{\text{Cl}}$ approaches and exceeds unity.

The vertical error bars in Fig. 3 are block-averaged standard errors of the mean that account for MD time correlations: we estimate the integrated autocorrelation time τ_{int} of $U_{\text{ex}}/(Nk_B T)$ from its normalized autocorrelation function, then partition the production trajectory into B non-overlapping, contiguous time segments & “blocks” of duration $L_{\text{blk}} \geq 10 \tau_{\text{int}}$. We then compute the block means $\{\bar{U}_b\}_{b=1}^B$ of $U_{\text{ex}}/(Nk_B T)$, and report the standard error of the mean

$$\text{SEM} = \frac{\text{std}(\{\bar{U}_b\})}{\sqrt{B}}. \quad (17)$$

The horizontal whiskers reflect uncertainty in ζ due to temperature sampling: we apply the same blocking procedure to the instantaneous temperature to obtain a blocked mean $\bar{T}$ and SEM_T , then propagate to ζ using

$$\frac{\delta\zeta}{\zeta} = \frac{1}{2} \frac{\text{SEM}_T}{\bar{T}} \Rightarrow \delta\zeta = \frac{1}{2} \zeta \frac{\text{SEM}_T}{\bar{T}}. \quad (18)$$

This procedure yields statistically independent uncertainty estimates and avoids underestimating errors due to time correlations. Here, the simulation temperature itself is fixed by the chosen inverse temperature β ; the “instantaneous temperature” is the kinetic estimator obtained from the bead velocities, which fluctuates around the target value in equilibrium. Because $\zeta \equiv b_Q/b_{\text{Cl}} \propto T^{-1/2}$, these natural thermal fluctuations induce only a very small statistical uncertainty in ζ , which we quantify and report for completeness rather than indicating any drift of the imposed temperature.

To establish the underlying equilibrium structure of the UEG across the full range of ζ , we computed the radial distribution function $g(r)$ for all cases (Fig. 4). At $\Gamma = 0.1$, the system is only weakly

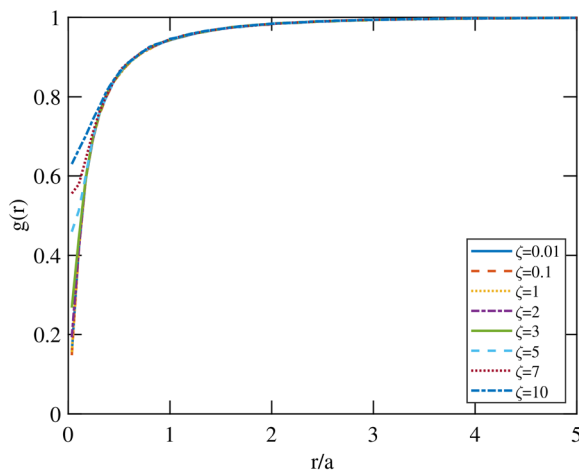


FIG. 4. Radial distribution function $g(r)$ for all ζ values at $\Gamma = 0.1$. All curves collapse for $r/a \geq 0.4$, reflecting the weakly correlated structure of the UEG across the full ζ range.

correlated, and all curves exhibit a narrow correlation hole at short distances followed by a smooth approach to unity, as expected for weakly coupled Coulomb fluids.^{55,56} For $r/a \geq 0.4$ the $g(r)$ curves essentially collapse, indicating that medium- and long-range structural correlations are governed by the weak Coulomb coupling and are, within our resolution, insensitive to ζ . Visible ζ -dependence is confined to the short-range correlation hole: as ζ increases, $g(r)$ in this region rises slightly, so the correlation hole becomes less deep. This behavior is consistent with quantum diffraction mildly softening the short-range repulsion between electrons, while leaving the overall screening structure effectively unchanged at the level resolved here.

To further assess the origin of the residual $\sim 1\%$ offset in the classical limit, we also performed sample simulations with an order of magnitude larger system size, increasing from $N = 5000$ to $N = 50\,000$ electrons for representative state points. In these large- N tests, the excess energy moved significantly closer to the analytical classical EOS, with deviations reduced to as small as 0.17%, albeit at substantially higher computational cost. This supports the interpretation that the remaining $\lesssim 1\%$ discrepancy in Fig. 3 is a benign finite-size effect rather than a deficiency of the RPMD sampling.

B. Diffusive transport

To assess transport in the UEG, we consider diffusion. We define the electron-wise centroid,

$$\mathbf{R}_i(t) = \frac{1}{P} \sum_{p=1}^P \mathbf{r}_{ip}(t), \quad (19)$$

unwrap $\mathbf{R}_i(t)$ across periodic boundaries and remove any residual system-wide center-of-mass drift. The multiple-time-origin MSD is

$$\text{MSD}(t) = \left\langle \frac{1}{N} \sum_{i=1}^N \|\mathbf{R}_i(t_0 + t) - \mathbf{R}_i(t_0)\|^2 \right\rangle_{t_0}, \quad (20)$$

which shows an initial ballistic t^2 rise followed by a long-time linear regime; representative curves are shown in Fig. 5.

All curves show the expected ballistic t^2 rise at very short times and a clear linear, diffusive regime at long times. At $\zeta = 0.1$ the $P = 1$ and $P = 10$ MSD traces are indistinguishable within noise (classical-like transport). For larger ζ the long-time slopes begin to separate: at $\zeta = 3$ and $\zeta = 7$, the RPMD curves exhibit a visibly steeper diffusive tail than $P = 1$, consistent with quantum-diffraction-enhanced diffusion as reported in Fig. 6. We therefore interpret the systematic, ζ -dependent separation between the MSD slopes for $P = 1$ and RPMD as the dynamical counterpart of the quantum corrections already quantified in our EOS analysis, rather than as numerical noise or a breakdown of diffusive behavior.

We can justify using the centroid MSD in place of the true bead average as follows. By writing each bead coordinate as a centroid plus an internal displacement, $\mathbf{r}_{ip}(t) = \mathbf{R}_i(t) + \mathbf{u}_{ip}(t)$ with $\frac{1}{P} \sum_{p=1}^P \mathbf{u}_{ip}(t) = \mathbf{0}$, the bead-averaged MSD decomposes as

$$\frac{1}{P} \sum_{p=1}^P \langle \|\Delta \mathbf{r}_{ip}(t)\|^2 \rangle = \langle \|\Delta \mathbf{R}_i(t)\|^2 \rangle + C_{\text{int}}(t), \quad (21)$$

where $\Delta \mathbf{r}_{ip}(t) \equiv \mathbf{r}_{ip}(t_0 + t) - \mathbf{r}_{ip}(t_0)$ and $\Delta \mathbf{R}_i(t) \equiv \mathbf{R}_i(t_0 + t) - \mathbf{R}_i(t_0)$. The internal contribution is

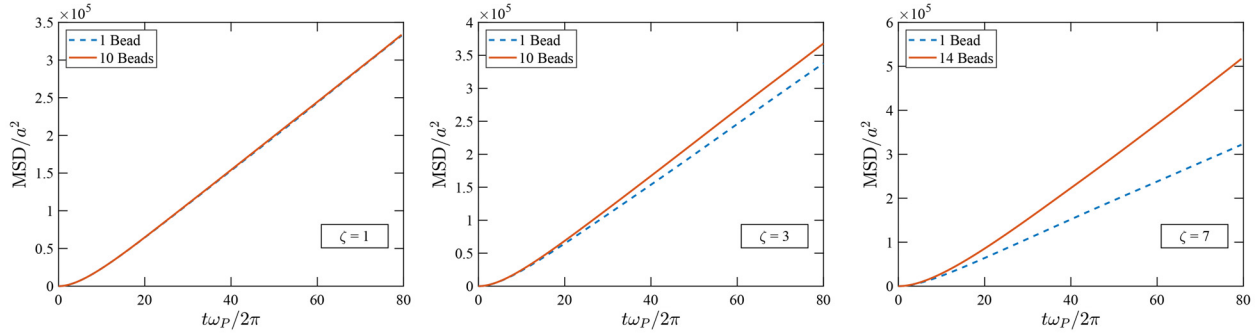


FIG. 5. Centroid mean-square displacement $\text{MSD}(t)$ at $\Gamma = 0.1$ for three quantum-classical ratios: $\zeta = 1$ (left), $\zeta = 3$ (middle), and $\zeta = 7$ (right). Axes are in reduced units: time is expressed in plasma periods $T_p \equiv 2\pi/\omega_p$ [abscissa $t/T_p = t\omega_p/(2\pi)$], and displacement is normalized by the Wigner-Seitz radius squared a^2 .

$$\begin{aligned} C_{\text{int}}(t) &\equiv \frac{1}{P} \sum_{p=1}^P \langle ||\mathbf{u}_{ip}(t) - \mathbf{u}_{ip}(0)||^2 \rangle \\ &= 2\langle u^2 \rangle - \frac{2}{P} \sum_{p=1}^P \langle \mathbf{u}_{ip}(t) \cdot \mathbf{u}_{ip}(0) \rangle, \end{aligned} \quad (22)$$

with stationarity implying $\langle ||\mathbf{u}_{ip}(t)||^2 \rangle = \langle ||\mathbf{u}_{ip}(0)||^2 \rangle \equiv \langle u^2 \rangle$, the mean-squared internal (bead-centroid) displacement. In RPMD, the internal ring-polymer modes are bounded harmonic normal modes [with frequencies $\omega_k = 2\omega_p \sin(k\pi/P)$] with $k = 1, \dots, P-1$. For a single mode, one has $\langle \mathbf{u}_{ik}(t) \cdot \mathbf{u}_{ik}(0) \rangle = \langle |\mathbf{u}_{ik}|^2 \rangle \cos(\omega_k t)$; summing over many modes, beads, particles, and time origins *dephases* these oscillations, so their average tends to zero on times long compared to the internal periods (weak anharmonicity damps them further). Hence, $C_{\text{int}}(t) \rightarrow 2\langle u^2 \rangle$, a time-independent plateau.^{10–12} For a free-particle ring polymer $\langle u^2 \rangle \propto \hbar^2 \beta / m = \mathcal{O}(\Lambda_{\text{th}}^2)$ (with an order-unity prefactor), so the plateau $2\langle u^2 \rangle$ is a constant vertical shift in the

MSD.^{12,19} In the present conditions, Λ_{th} is negligible compared with the root-mean-squared centroid displacement over the diffusive fitting window $\sqrt{6Dt}$, and the MSD scale is $6Dt$; thus $2\langle u^2 \rangle = \mathcal{O}(\Lambda_{\text{th}}^2)$ is small relative to $6Dt$ and does not affect the long-time slope of the MSD. Consequently the bead-averaged and centroid MSDs share the same long-time linear growth, and their slopes—hence the diffusion coefficient—are nearly identical; the fractional impact of the internal term on a slope taken at time t scales as $2\langle u^2 \rangle / (6Dt) = \mathcal{O}(\Lambda_{\text{th}}^2 / (Dt)) \rightarrow 0$.

Einstein's relation connects the long-time slope to diffusion,⁵⁷

$$D = \lim_{t \rightarrow \infty} \frac{\text{MSD}(t)}{6t}. \quad (23)$$

Together with (20), we extract D by linear regression over a long-time diffusive window. Operationally, we identify the diffusive window by monitoring the instantaneous slope of $\text{MSD}(t)$ and selecting the earliest contiguous interval over which the slope has reached its asymptotic value and remains within $\sim 10\%$ of its median for at least one decade in t . This criterion robustly detects the onset of the linear, long-time regime and avoids bias from the short-time ballistic curvature. The statistical uncertainty of the slope is estimated by resampling the time-origin ensemble with block lengths $\geq 10 \tau_{\text{int}}$, where τ_{int} is the integrated autocorrelation time of the per-lag MSD series. We report the dimensionless diffusivity,

$$D^* = \frac{D}{a^2 \omega_p}. \quad (24)$$

We work at fixed density, so a and ω_p are constant; hence any variation in D^* reflects only the temperature dependence of $D(T)$. In weak-coupling kinetic theory, one has $D \propto T^{5/2} / (n \ln \Lambda)$ at fixed n ;^{9,58,59} neglecting the weak T -dependence of $\ln \Lambda$ for error propagation, we use

$$\frac{\delta D^*}{D^*} = \frac{5}{2} \frac{\text{SEM}_T}{\bar{T}}, \quad (25)$$

where SEM_T is the block-averaged standard error of the mean temperature and $\bar{T}$ the corresponding block-averaged mean.

For theoretical context and comparison, we evaluate the UEG Chapman-Enskog (CE) self-diffusion. The first-Sonine CE result is

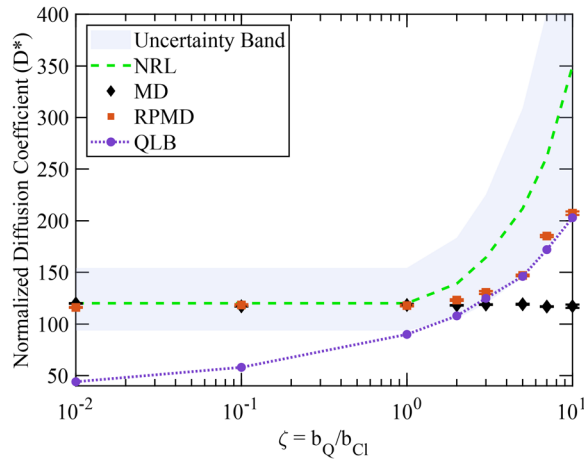


FIG. 6. Dimensionless diffusion coefficient $D^* = D/(a^2 \omega_p)$ vs $\zeta = b_Q/b_{\text{Cl}}$ at $\Gamma = 0.1$. Black diamonds: classical MD ($P = 1$); red squares: RPMD ($P = 10 - 18$). Purple dashed line: quantum Lenard-Balescu (QLB) prediction. The green dashed line denotes the NRL Chapman-Enskog prediction and the gray band its corresponding $\mathcal{O}(1/\ln \Lambda)$ uncertainty.

$$[D^*]_1 = \frac{\sqrt{\pi/3}}{\Gamma^{5/2}} \frac{1}{\Xi_{11}}, \quad (26)$$

and the standard second-Sonine correction gives

$$D_{\text{CE}}^* = \frac{[D^*]_1}{1 - \Delta}, \quad (27)$$

$$\Delta = \frac{(2\Xi_{12} - 5\Xi_{11})^2}{\Xi_{11}(55\Xi_{11} - 20\Xi_{12} + 4\Xi_{13} + 8\Xi_{22})}. \quad (28)$$

(See Chapman and Cowling,⁶⁰ Braginskii,⁵⁹ the specific UEG forms above follow Eqs. (39)–(41) of Baalrud and Daligault.⁷)

In weak coupling, the generalized Coulomb integrals Ξ_{lk} reduce to the Coulomb logarithm plus order-unity corrections. With $\Psi(x) \equiv \Gamma'(x)/\Gamma(x)$ the digamma function and $\Psi(1) = -\gamma_E$, define $\lambda \equiv \ln \Lambda - \gamma_E$. Then the working offsets are

$$\begin{aligned} \Xi_{11} &= \lambda, & \Xi_{12} &= \lambda + 1, \\ \Xi_{13} &= 2\left(\lambda + \frac{3}{2}\right), & \Xi_{22} &= 2\left(\lambda + \frac{1}{2}\right), \end{aligned} \quad (29)$$

consistent with the weak-coupling expansion [cf. Eqs. (34a)–(34c) in Ref. 7]

For the electron–electron Coulomb logarithm, we adopt the *NRL Plasma Formulary*'s nonrelativistic expression (with n in cm^{-3} , T in eV),

$$\ln \Lambda_{ee} = 23.5 - \frac{1}{2} \ln n + \frac{5}{4} \ln T - \sqrt{10^{-5} + \frac{(\ln T - 2)^2}{16}}, \quad (30)$$

which, when inserted into (26)–(29) via $\lambda = \ln \Lambda_{ee} - \gamma_E$, provides the analytic prediction $D_{\text{NRL}}^* \equiv D_{\text{CE}}^*[\ln \Lambda_{ee}]$.^{7,9,59–61}

The *NRL Formulary* notes that weak-coupling transport truncation errors scale as $\mathcal{O}(1/\ln \Lambda)$.⁹ We represent this using a log-symmetric multiplicative band with half-width,

$$\delta \equiv \frac{1}{\ln \Lambda_{ee}}, \quad (31)$$

so that

$$D_{\text{upper}}^* = D_{\text{NRL}}^* (1 + \delta), \quad D_{\text{lower}}^* = \frac{D_{\text{NRL}}^*}{1 + \delta}. \quad (32)$$

This implements the stated $\mathcal{O}(1/\ln \Lambda)$ behavior and is symmetric in $\ln D^*$, which avoids bias when relative errors are large. For the present cases the one-sided relative uncertainty increases from $\sim 28\%$ at $\zeta \leq 1$ ($\ln \Lambda_{ee} = 3.54$) to $\sim 66\%$ at $\zeta = 10$ ($\ln \Lambda_{ee} = 1.51$).

Figure 6 overlays our RPMD data on the NRL analytic curve and uncertainty band (32). Both $P = 1$ and $P = 10 - 14$ agree with the theory up to $\zeta \lesssim 1$. Across the crossover, the RPMD diffusivity rises more gradually than the NRL curve. By contrast, $P = 1$ values remain nearly flat, consistent with classical (point-particle) dynamics that omit diffraction. Because the NRL band widens to tens of percent for $\ln \Lambda_{ee} \lesssim 2$, it becomes insufficiently predictive in the crossover. The RPMD trajectories therefore provide a sharper, dynamics-based constraint on $D^*(\zeta)$ and highlight diffractive softening beyond what the NRL Coulomb-log-based kinetic prediction captures in this regime.

To further constrain the diffusivity in the quantum-classical crossover, we compare our results with a higher-fidelity kinetic theory

TABLE III. Comparison of dimensionless diffusivities D^* obtained from the NRL Coulomb–logarithm model, RPMD simulations, and quantum Lenard–Balescu (QLB) kinetic theory across the $\zeta = b_Q/b_{\text{Cl}}$ grid.

ζ	NRL	RPMD	QLB
0.01	120.25	116.17	44
0.10	120.25	118.55	58
1	120.25	117.97	90
2	138.93	123.32	108
3	164.39	130.53	125
5	211.95	146.93	146
7	261.68	185.20	172
10	349.85	207.41	203

that does not rely on a prescribed Coulomb logarithm. For this purpose we include quantum Lenard–Balescu (QLB) predictions computed using the conservative spectral solver of Scullard *et al.*⁶² and Whitley *et al.*⁶³ The QLB equation incorporates quantum diffraction and dynamical screening self-consistently, so the logarithmic and order-unity terms in the collision operator arise naturally from the theory rather than by external parameterization.

The QLB diffusivities exhibit a quantitative agreement once $\zeta \geq 2$, precisely where quantum diffraction begins to dominate the close-impact scattering physics that controls the diffusion coefficient. In this regime, where quantum effects become dominant, the NRL Chapman–Enskog prediction loses accuracy due to its reliance on a single Coulomb logarithm, whereas both QLB and RPMD capture the diffractive softening of close encounters. The close agreement between QLB and RPMD therefore provides a parameter-free, first-principles benchmark supporting the accuracy of the RPMD diffusivities in the quantum-classical crossover.

The full set of diffusivities from NRL, RPMD, and QLB is listed in Table III, which summarizes the agreement patterns described above.

C. Scattering

Textbook small-angle Coulomb scattering defines the Coulomb logarithm by impact-parameter cutoffs, $\ln \Lambda = \ln(b_{\text{max}}/b_{\text{min}})$.⁸ In the classical treatment one takes $b_{\text{max}} \simeq \lambda_D$ (Debye screening) and $b_{\text{min}} \simeq b_0$ (classical closest approach), which in the limit $b_{\text{max}} \gg b_0$ gives $\ln \Lambda \approx \ln(\lambda_D/b_0)$.⁸ Under our conditions, the *short-range* cutoff can be set by quantum diffraction: when the thermal de Broglie wavelength $b_Q \equiv \lambda_{\text{th}}$ exceeds the classical scale, the appropriate choice is $b_{\text{min}} = b_Q$; more generally for Maxwellian transport $b_{\text{min}} = \max\{b_{\text{Cl}}, b_Q\}$. Here, $b_{\text{Cl}}(T) \equiv e^2/(4\pi\epsilon_0 k_B T)$ is the thermal counterpart of the textbook $b_0(V) = e^2/(4\pi\epsilon_0 \mu V^2)$ (i.e., b_0 evaluated at a thermal speed, up to an $\mathcal{O}(1)$ factor). We parameterize the classical–quantum crossover by $\zeta \equiv b_Q/b_{\text{Cl}}$; thus $\zeta \ll 1$ recovers Rutherford-like behavior, while $\zeta \gtrsim 1$ is diffraction-dominated.

To test the hypothesis that quantum scattering “softens” binary collisions, we hold T fixed (hence b_{Cl} and b_Q fixed) and vary the actual impact parameter b , examining three kinematic regimes $b > b_Q$, $b = b_Q$, and $b < b_Q$ to isolate diffraction effects relative to the classical benchmark.

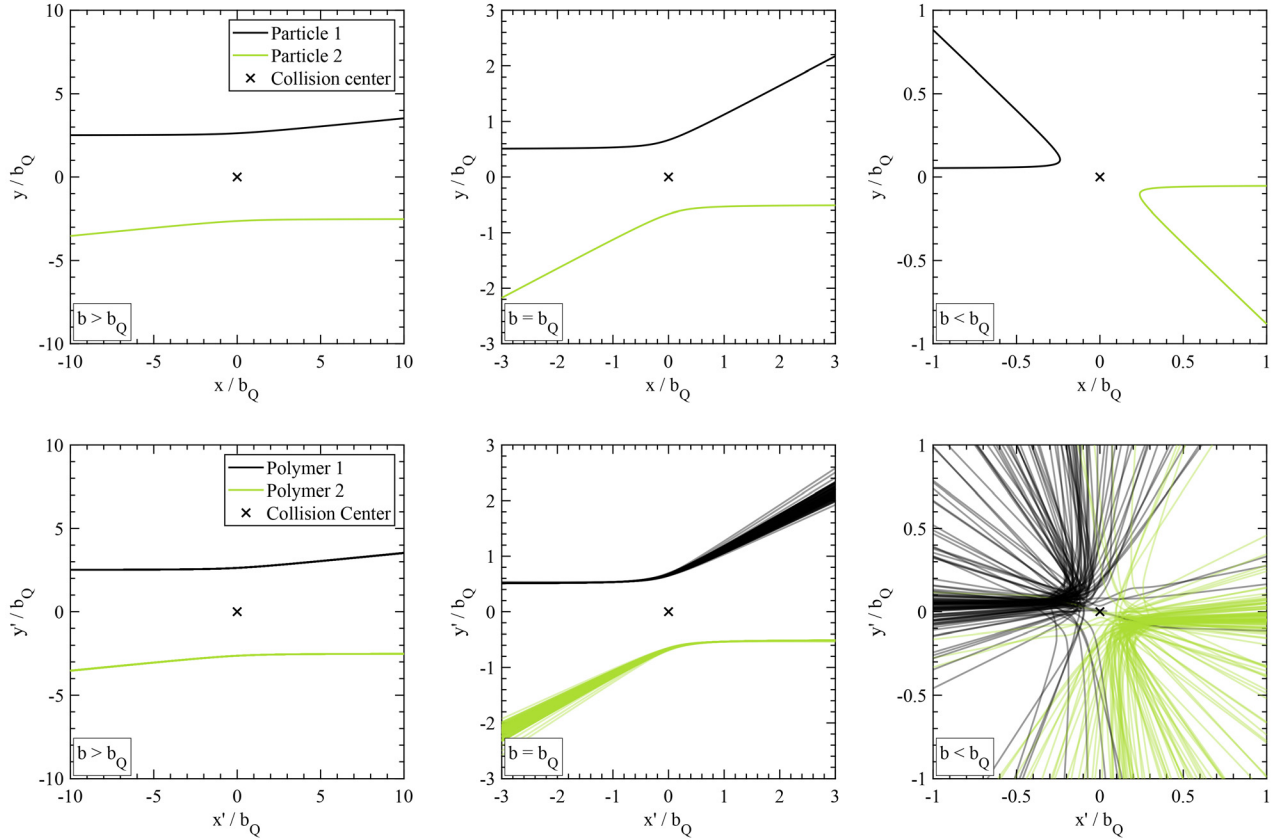


FIG. 7. Centroid trajectories for two-electron collisions in the three impact-parameter regimes: (i) $b > b_Q$, (ii) $b = b_Q$, and (iii) $b < b_Q$. Top row: single classical point-particle scattering shown in the lab frame $(x, y)/b_Q$. Bottom row: ensemble of $N = 100$ ring-polymer centroid trajectories, displayed in primed axes $(x', y')/b_Q$ after a rigid Rodrigues rotation that aligns each case to the scattering plane. The rotation is for visualization only; all dynamics and angles are computed from the unrotated velocities. All panels share identical spatial scaling.

We first characterized the internal fluctuations, i.e., per-bead deviations from the centroid within a single ring polymer, from independent RPMD runs at the same (n, T) as our scattering simulations, accumulating the empirical distributions of bead-centroid displacements and velocity offsets: $\Delta \mathbf{r}_{ip} \equiv \mathbf{r}_{ip} - \mathbf{R}_i$ and $\Delta \mathbf{v}_{ip} \equiv \mathbf{v}_{ip} - \mathbf{V}_i$, with $\mathbf{R}_i = \frac{1}{P} \sum_p \mathbf{r}_{ip}$ and $\mathbf{V}_i = \frac{1}{P} \sum_p \mathbf{v}_{ip}$. Initial ring-polymer configurations were then constructed by sampling from these distributions (randomizing directions for isotropy) and re-centering each polymer to enforce $\sum_p \Delta \mathbf{r}_{ip} = \mathbf{0}$ and $\sum_p \Delta \mathbf{v}_{ip} = \mathbf{0}$. We then define the scattering angle from the *centroid* trajectory, which provides a low-noise, physically comparable deflection measure, while the bead spread encodes quantum dispersion/diffraction.

To impose a given impact parameter b , we initialize the two ring-polymer centroids at $\mathbf{C}_1(0) = (-x_0, +b/2, 0)$ and $\mathbf{C}_2(0) = (+x_0, -b/2, 0)$ with velocities $(+v_{th}, 0, 0)$ and $(-v_{th}, 0, 0)$, respectively. After sampling bead fluctuations, a centroid lock is applied to enforce $|\mathbf{C}_{1,y}(0) - \mathbf{C}_{2,y}(0)| = b$, so the transverse separation equals the prescribed impact parameter.

For each regime, we run an ensemble of $N = 500$ ring-polymer scatterings and, for direct comparison, a single classical two-body collision. Typical centroid trajectories ($N = 100$) are shown in Fig. 7; Table IV reports the mean centroid angle θ_{RP} alongside the classical

point-particle angle θ . For visualization, centroid trajectories are rotated to the scattering plane (set $\phi \rightarrow 0$) using Rodrigues' rotation formula;^{64,65} angles are computed from the unrotated velocities and are unaffected by this step.

When $b > b_Q$, the ring-polymer and point-particle results coincide ($\theta_{RP} \approx \theta \approx 5.72^\circ$) with negligible spread. At $b = b_Q$, the polymer ensemble broadens ($\theta_{RP} = 27.73^\circ \pm 1.93^\circ$), reflecting quantum uncertainty. Below the quantum scale ($b < b_Q$), the mean polymer angle ($\theta_{RP} \approx 113.79^\circ$) is about 16.5% smaller than the classical value ($\theta = 136.34^\circ$), demonstrating the “softening” of collisions by quantum diffraction. Angular quantities are summarized using *circular* statistics, where σ denotes the circular standard deviation (shot-to-shot

TABLE IV. Centroid scattering angles θ_{RP} for ring-polymer ensembles (mean $\pm 1\sigma$) vs classical point-particle angle θ ($N = 500$ per regime).

Regime	Point-particle θ ($^\circ$)	Ring-polymer θ_{RP} ($^\circ$)
$b = 5b_Q$	5.72	5.72 ± 0.02
$b = b_Q$	28.03	27.73 ± 1.93
$b = 0.1b_Q$	136.34	113.79 ± 48.18

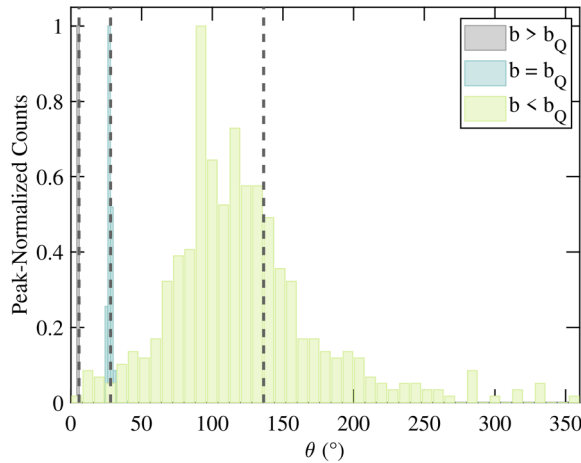


FIG. 8. Histograms of the ring-polymer centroid scattering angle θ_{RP} for three impact-parameter regimes: gray ($b > b_Q$), blue ($b = b_Q$), and green ($b < b_Q$). Each histogram aggregates all samples from the $N = 500$ ensemble and is peak-normalized to unity to enable direct comparison of shape and spread. Dashed vertical lines mark the corresponding classical point-particle scattering angle θ . Bin widths are case-specific (coarser for $b < b_Q$) to mitigate sparse bins.

spread). Figure 8 shows histograms of θ_{RP} for each regime, peak-normalized to unit height. Dashed vertical lines mark the corresponding classical angle θ , enabling direct comparison of angular spread and mean shifts.

In RPMD, each electron is a ring polymer whose bead cloud has thermal size $b_Q \sim \Lambda_{th}$. For $b > b_Q$ the cloud is small compared to the encounter scale, so all beads sample nearly the same field and the centroid follows the classical deflection (bottom-left panel of Fig. 7), yielding a narrow θ_{RP} distribution. As b approaches and drops below b_Q , different beads probe different parts of the Coulomb field at closest approach; the centroid obeys $m \ddot{\mathbf{R}} = (1/P) \sum_p \mathbf{F}(\mathbf{R} + \mathbf{u}_p)$, i.e., a *bead-averaged* force. This bead averaging effectively corresponds to convolving the point-Coulomb potential with an isotropic Gaussian kernel of standard deviation σ (variance $\sigma^2 = \langle u^2 \rangle \sim \Lambda_{th}^2$), replacing $1/r$ by $\text{erf}(r/(\sqrt{2}\sigma))/r$ and thereby smoothing the short-range singularity. The net result is (i) a reduced *mean* deflection ($\theta_{RP} < \theta$) and (ii) a larger *spread* of outcomes (broader θ_{RP} histograms)—diffraction-like behavior encoded by the ring-polymer springs (see Fig. 8). Because our conditions are non-degenerate ($n\Lambda_{th}^3 \ll 1$), exchange and many-body interference are negligible; the dominant quantum effect is this diffraction/zero-point broadening.

IV. SUMMARY AND OUTLOOK

Within the weakly coupled, non-degenerate regime explored here ($\Gamma = 0.1$, $n\Lambda_{th}^3 \ll 1$), RPMD reproduces the weak-coupling UEG equation of state for static observables and captures the classical-to-quantum transport trend. The dimensionless diffusivity $D^*(\zeta)$ increases with ζ : for $\zeta \lesssim 1$ both $P = 1$ and $P = 10$ agree with the Chapman–Enskog prediction evaluated with the *NRL Formulary* Coulomb logarithm, whereas across the crossover the $P = 10 - 18$ curve rises *more gradually* than the analytic trend; by contrast, $P = 1$ remains nearly flat, consistent with classical dynamics that omit diffraction. Because the *NRL* band widens to tens of percent as $\ln \Lambda$

decreases, the RPMD trajectories provide a sharper, dynamics-based constraint on $D^*(\zeta)$ in this regime.

Independent validation is provided by quantum Lenard–Balescu (QLB) diffusivities, which exhibit quantitative agreement with RPMD for $\zeta \geq 2$, precisely where quantum diffraction governs close-impact scattering. This parameter-free kinetic benchmark confirms that RPMD captures the correct quantum–classical crossover beyond the range where a single Coulomb logarithm remains predictive. Together with the ensemble scattering analysis (Table IV, Fig. 7), these observations give a coherent microscopic-to-macroscopic picture: As collisions probe scales $b \lesssim b_Q$, diffraction softens binary deflections ($\theta_{RP} < \theta$) and, in aggregate, suppresses electronic self-diffusion.

RPMD is exact for equilibrium statistics in the $P \rightarrow \infty$ limit and provides a pragmatic approximation to Kubo-transformed real-time correlations by evolving the ring-polymer dynamics.^{10,11} In the present setting—where DFT+KG supplies mean-field conductivities^{38,39,44} and PIMC excels at static properties but not direct real time¹⁹—RPMD offers a complementary, fully interacting, trajectory-based route to electronic self-diffusion (transport).

Our conclusions apply within clearly stated bounds: weak coupling and non-degeneracy (Boltzmann electrons). Earlier sections showed that centroid- and all bead-based MSDs/diffusivities are statistically indistinguishable in our data, so we report centroid results for simplicity.^{10,11} We verified bead-number convergence of static observables; accordingly, we used $P = 10$ for $\zeta = 0.1 - 3$, $P = 14$ for $\zeta = 5$ and 7, and $P = 18$ for $\zeta = 10$. Residual $\mathcal{O}(1/P^2)$ Trotter errors were negligible compared with statistical uncertainties.¹²

Here electrons obey Boltzmann statistics (to avoid the fermionic sign problem), so statements about $\Theta \lesssim 1$ are outside of the scope.¹⁹ A promising path forward is to embed Fermi–Dirac statistics via the continuous-statistics PIMD framework of Xiong and Xiong, which fits thermodynamic observables $E(\beta, \zeta)$ for $\zeta \in [0, 1]$ (bosons/distinguishable) and analytically continues to $\zeta = -1$ (fermions), thereby providing a controlled route past the non-degenerate regime.²³

The next step is to extend from the electron UEG to *multi-component plasmas* (e.g., hydrogenic mixtures) to investigate transport properties of realistic neutral plasmas, rather than the UEG model system. This will enable direct comparisons to crossover-aware kinetic theories (e.g., BPS and effective-potential treatments) across coupling and degeneracy,^{6,7} and sets the stage for introducing Fermi–Dirac statistics via the ζ -extrapolation PIMD route.^{23,66,67}

Looking ahead, extending RPMD to multi-component plasmas will enable the study of a broader class of dynamical transport properties relevant to warm dense and inertial-fusion plasmas. Because RPMD provides approximate real-time Kubo-transformed correlation functions in the low-frequency, near-equilibrium regime, it can be applied to investigate electron–ion momentum exchange, electrical and thermal conductivities, and the low-frequency limits of collective density fluctuations that underlie the dynamical structure factor $S(k, \omega)$. Applying RPMD in such settings would offer a fully interacting, trajectory-based benchmark for assessing crossover-aware kinetic theories and for clarifying how quantum diffraction and species mass asymmetry shape transport in realistic plasmas.

In summary, our results indicate that RPMD—designed to be exact for equilibrium statistics—also provides a practical, quantitatively consistent framework for probing low-frequency, near-equilibrium

transport in weakly coupled electron plasmas, including the classical-to-quantum crossover where widely used methods begin to lose accuracy. This motivates its use as a reference tool for developing crossover-aware transport models and for extending electron-only studies to realistic multi-component plasmas.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

I. Pagiaitis: Conceptualization (supporting); Formal analysis (lead); Investigation (lead); Methodology (equal); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **N. R. Shaffer:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **S. X. Hu:** Conceptualization (supporting); Project administration (lead); Resources (lead); Supervision (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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